

Cross-Sectional Variation of Risk-targeting Option Portfolios

Liuren Wu,^{*} Yaofei Xu,[†]

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Abstract

Option contracts are listed on thousands of stocks with different number of contracts per each name. This paper proposes to construct four risk-targeting option portfolios to consolidate the information in all the option contracts on each stock at any given date along four risk dimensions. A cross-sectional regression identifies the market price of each risk dimension. The pricing estimates positively predict the excess returns of the corresponding option portfolio. Long-short portfolio of option portfolio construction along each risk dimension in proportion to the market price of risk estimates generates highly positive risk-adjusted excess returns across all four risk dimensions.

JEL Classification: C13; C51; G12; G13

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^{*}Zicklin School of Business, Baruch College, The City University of New York, One Bernard Baruch Way, New York, NY 10010, USA. E-mail: liuren.wu@baruch.cuny.edu.

[†]School of Mathematics and Physics, Research Institute of Quantitative Finance, Xi'an Jiaotong Liverpool University, Wuzhong District, Suzhou, China. E-mail: Yaofei.Xu@xjtlu.edu.cn

In the U.S., several options exchanges list option contracts on thousands of stocks. For each stock, the exchanges list option contracts across a wide range of strike prices and expiration dates. The number of listed option contracts per stock differs greatly across different names. How to summarize the option behavior and make investment decisions across both the large stock universe and the disparate number of option contracts on different stocks remains a challenging task.

This paper proposes to summarize the option behavior on each stock by constructing four risk-targeting option portfolios. Each portfolio targets a unit risk along one risk dimension while neutral to other risk dimensions. The portfolio construction acts as a dimension reduction operation and consolidates information in all the option contracts along four risk dimensions, allowing us to readily compare the cross-sectional pricing and return variation along each risk dimension across the large stock universe.

Following the recent development in top-down decentralized option pricing (Carr and Wu (2020) and Wu and Zhang (2024)), this paper proposes to summarize the risk exposures of each option contract through the Black and Scholes (1973) and Merton (1973b) (henceforth BMS) value representation. The representation decentralizes the risk exposure estimation down to each contract, and represents the risk sources in observable variables, e.g., variations in the stock price and the contract's own BMS implied volatility. Through this representation, the delta-hedged excess return of an option contract can be attributed to variations in four risk dimensions: (i) the vega exposure capturing the option return contribution from the directional implied volatility movement, (ii) the gamma exposure capturing the contribution from the second-order stock price movement, (iii) the volga exposure capturing the contribution from the second-order implied volatility movement, and (iv) the vanna exposure capturing the contribution from the co-movement of the stock price and the option implied volatility. Based on the observed histories of stock returns and implied volatility movements, we construct historical variance and covariance estimators to proxy the cor-

responding statistical conditional moment forecasts. Multiplying the four sources of risk exposures on the option contracts by the corresponding statistical moment forecasts creates a risk-adjusted exposure matrix, from which we can construct four risk-targeting option portfolios to summarize the option behaviors along the four risk dimensions.

We perform empirical analysis on the U.S. stock options market over a period of 299 months from 1997 to 2021. We sample the data once a month, and construct a data sample that includes a total of 16,947,489 option contracts on 3,691 individual stocks. At each sampling date t and for each stock i in the universe, we construct the $(n_{t,i} \times 4)$ risk-adjusted exposure matrix for the $n_{t,i}$ selected option contracts on the stock. Regressing the options time decay on the risk-adjusted exposure matrix identifies the market prices of risk on the four risk dimensions. The regression R^2 estimates vary across the stock universe from an average of 96.7% to an average of 99.8% within the 10-90th percentile range, with an average median value of 99.2%. The high R^2 estimates show that the four risk-targeting option portfolios can effectively summarize almost all the pricing information in the different numbers of option contracts on each stock at any given date.

In parallel to the market price of risk estimates, we construct the delta-hedged excess returns over the next month on the same risk-targeting option portfolios. We use the same risk-adjusted risk exposure matrix, together with the pricing error from the pricing regression, to explain the cross-sectional option excess return variation over the next month. The risk structure also explains the option excess return variation well. The R^2 estimates from the forecasting return regression vary across the stock universe from an average of 52.9% to an average of 95.7% within the 10th-90th percentile range, with an average median value of 83.7%.

The return regression includes the pricing error from the pricing regression in addition to the risk-adjusted exposure matrix. The slope coefficient on the pricing error represents the excess re-

turn on a value-investing statistical arbitrage portfolio that targets a unit exposure to the option pricing error while being neutral to all four risk dimensions. The portfolio generates predominantly positive excess returns. An equally-weighted portfolio of the statistical arbitrage portfolio across the stock universe generates extremely high risk-adjusted average excess return. Therefore, the small remaining pricing variation unexplained by the four risk-targeting portfolios represents highly reverting mispricing and potential statistical arbitrage opportunities.

The return coefficients on the risk-adjusted exposure matrix represent the excess returns on the four risk-targeting option portfolios. Both the market price of risk estimates and the ex post excess returns on the four portfolios vary strongly across the stock universe. Under our theoretical setting, the market price of risk represents the conditional expectation of the future excess return on the corresponding risk-targeting portfolio. We examine the validity of the expectation hypothesis through a cross-sectional forecasting regression at each date on each risk source. We regress the ex post excess return of the risk-targeting portfolio over the next month against the ex ante market price of risk estimate cross-sectionally over the stock universe. In line with the expectation hypothesis, the regressions generate strongly positive and highly statistically significant average slope estimates on all four risk-targeting portfolios. Meanwhile, the average slope estimates are all below the null value of one and some of the average intercept estimates deviate from zero. These deviations from the expectation hypothesis can be driven by estimation errors in the market price of risk estimates and potential deficiencies of the theoretical framework.

Finally, we propose to construct long-short portfolios of each risk-targeting option portfolio in proportion to the corresponding market price of risk estimates. This long-short portfolio of portfolio construction does not involve any forecasting regressions; nor does it have any forward looking bias. The portfolio is constructed purely based on the market price of risk estimates obtained from a cross-sectional pricing regression on that date. The strategy generates strongly positive average

excess returns with high information ratios across all four risk dimensions, with or without taking on aggregate market exposures. As such, the market price of risk estimates offer a completely new dimension absent from existing literature, with which investors can not only perform market timing investment in any of the four risk dimensions in aggregate, but they can also benefit from the large cross-sectional market pricing variation across the stock universe at each date.

1. Literature

A rapidly growing literature examines stock option return behaviors and proposes empirical option return factor models.¹ Compared to this literature, our framework for constructing risk-targeting option portfolios allows us to summarize the option pricing and return behavior across the full spectrum of strikes and maturities. Without using risk-targeting option portfolio construction to summarize the information, the literature often relies on picking a single option contract or combination and compare their cross-sectional variation over the stock universe. Such an approach only picks up part of the information in the options market while discarding much of the rest.

More importantly, our framework offers a completely new dimension absent from existing empirical return factor models. Under our theoretical setting, regressing the options time decay on the risk-adjusted option exposures for each stock at each date generates the market price of risk estimates on the four risk sources for that stock. The market price of risk estimates positively predict the cross-sectional variation of the subsequently realized excess returns on the corresponding risk-targeting option portfolios in line with the expectation hypothesis. This allows us to construct

¹Prominent examples include Figlewski and Freund (1994), Coval and Shumway (2001), Bakshi, Kapadia, and Madan (2003), Goyal and Saretto (2009), Cao and Han (2013), Boyer and Vorkink (2014), Heston, Jones, Khorram, Li, and Mo (2021), Zhan, Han, Cao, and Tong (2022), Bali, Beckmeyer, Möerke, and Weigert (2023), Horenstein, Vasquez, and Xiao (2023), Tian and Wu (2023), Vasquez and Xiao (2023), and Käfer, Mörke, Weigert, and Wiest (2024).

highly profitable portfolio of option portfolios along each risk dimension based on the market price of risk estimates, without relying on rolling estimation of any return forecasting regressions.

When researchers propose investment strategies based on empirical return factor models, the strategy necessarily relies on historical rolling estimation of the return forecasting regressions. The general strategy is to be long a factor portfolio if the rolling historical average of the factor portfolio excess returns is significantly positive, and short if the historical average excess return is significantly negative. As is well-documented and frequently cautioned against, historical performance does not necessarily carry into the future. This is especially true when the risk premium on a risk dimension varies strongly over time and switches signs. By contrast, strong time variation in the risk premium of a risk dimension does not reduce the efficacy of our strategy, but rather makes it all the more useful relative to the traditional strategy relying on historical averages.

Furthermore, the average excess return on a factor portfolio represents the average risk premium for a unit aggregate market exposure to the underlying risk source; yet our estimation shows that the market pricing on each risk dimension shows large cross-sectional variation over the stock universe. The aggregate market risk premium is small relative to the large cross-sectional variation. Our strategy does not rely on the significance of the average aggregate risk premium of a risk dimension, but directly benefits from the large cross-sectional variation. In fact, we can construct the portfolio of option portfolios without taking on any aggregate exposure to the risk source. Removing the aggregate exposure significantly reduces the return volatility and greatly enhances the risk-adjusted performance of the long-short portfolio of option portfolios.

As the number of option contracts can vary greatly across different stocks, constructing risk-targeting option portfolios allows us to summarize the information from all the option contracts along the major risk dimensions. We perform dimension reduction and portfolio construction

based on the top-down decentralized BMS value representation. In principle, one can also perform dimension reduction based on classic bottom-up option pricing models. These models generate option valuation by specifying the risk-neutral dynamics of the underlying security price and taking expectation on the payoffs of the different option contracts based on the same underlying dynamics.² Nevertheless, due to the complex nature of bottom-up modeling and identification constraints, the common designs of bottom-up option pricing models in the literature are often confined to low-dimensional specifications. The low dimensionality severely limits their explanatory powers on the option pricing behavior and the option return variation (Israelov and Kelly (2017) and Wu and Zhang (2024)).

2. Theoretical background

This paper proposes to construct risk-targeting option portfolios to summarize the behaviors of the different number of option contracts underlying each stock.

2.1. Decentralized risk attribution and option pricing

The top-down decentralized option pricing framework by Carr and Wu (2020) and Wu and Zhang (2024) is particularly suited for this purpose. The framework starts with a market with riskfree bonds, a stock, and a set of option contracts written on the stock. For notational simplicity, we represent the values of the stock and its option contracts in their corresponding forward space. The returns on these forward securities are essentially excess returns over the riskfree bonds. We assume no dynamic arbitrage between each option contract and the stock, so that there exists a

²Prominent examples include the jump-diffusion model of Merton (1976), the stochastic volatility model of Heston (1993), and more sophisticated specifications than accommodate both features, e.g., Bates (1996), Bakshi, Cao, and Chen (1997), Duffie, Pan, and Singleton (2000), and Carr and Wu (2004, 2007a, 2007b, 2010, 2011).

risk-neutral probability measure $\mathbb{Q}$, equivalent to the statistical probability measure $\mathbb{P}$, such that the expected excess returns on the stock and the option contracts are zero.

The framework summarizes the risk exposures of an option contract j on an underlying stock i via the BMS pricing equation, $B(t, S_t, I_t)$, in terms of the calendar time t , the underlying stock forward price S_t , and the option's implied volatility I_t . Through the BMS representation, we highlight the risk exposures of the option contract by attributing its forward value variation (i.e., its excess profit and loss) via a Taylor expansion of the value function against the three variates,

$$dB = B_t dt + B_S S_t \frac{dS_t}{S_t} + B_I I_t \frac{dI_t}{I_t} + \frac{1}{2} B_{SS} S_t^2 \left(\frac{dS_t}{S_t} \right)^2 + \frac{1}{2} B_{II} I_t^2 \left(\frac{dI_t}{I_t} \right)^2 + B_{IS} I_t S_t \left(\frac{dS_t}{S_t} \frac{dI_t}{I_t} \right), \quad (1)$$

where we adopt the continuous-time notation and use (dB, dt, dS_t, dI_t) to denote the short-term changes in the forward option value, the calendar time, the stock forward price, and the contract's implied volatility, respectively. The attribution is applicable to any option contract j on any underlying stock i . To reduce notation cluttering, we drop the stock-option contract (i, j) reference when no confusion shall occur. We also suppress the arguments, (t, S_t, I_t) , of the value function and use $B_t, B_S, B_I, B_{SS}, B_{II}, B_{IS}$ to denote the partial derivatives.

Equation (1) performs the expansion to the second order while ignoring higher-order effects. The first term captures the option's time decay through the option's theta exposure (B_t). As time moves deterministically, the attributed option variation from time decay is deterministic. The second term captures the option's delta exposure (B_S). Through delta hedge, as is often required for institutional investors, one can remove the delta exposure of the option contract. The remaining terms capture the risk exposures of the delta-hedged option investment. They are commonly known as the option's cash vega ($B_I I_t$), cash gamma ($B_{SS} S_t^2$), cash volga ($B_{II} I_t^2$), and cash vanna ($B_{IS} I_t S_t$).

Taking expectation under the risk-neutral measure $\mathbb{Q}$ on the attribution in (1) and imposing the

no-dynamic-arbitrage condition between the option contract and the underlying security, Carr and Wu (2020) derive the decentralized pricing relation on the option contract,

$$-B_t = B_{II_t}\mu_t + \frac{1}{2}B_{SS}S_t^2v_t + \frac{1}{2}B_{II}I_t^2\omega_t + B_{IS}I_tS_t\gamma_t, \quad (2)$$

where $(\mu_t, v_t, \omega_t, \gamma_t)$ denote the risk-neutral conditional moment conditions on the security forward return and the proportional implied volatility change,

$$\mu_t \equiv \frac{1}{dt}\mathbb{E}_t\left[\frac{dI_t}{I_t}\right], \quad v_t \equiv \frac{1}{dt}\mathbb{E}_t\left[\left(\frac{dS_t}{S_t}\right)^2\right], \quad \omega_t \equiv \frac{1}{dt}\mathbb{E}_t\left[\left(\frac{dI_t}{I_t}\right)^2\right], \quad \gamma_t \equiv \frac{1}{dt}\mathbb{E}_t\left[\frac{dS_t}{S_t}, \frac{dI_t}{I_t}\right], \quad (3)$$

with $\mathbb{E}_t[\cdot]$ denoting the expectation under the $\mathbb{Q}$ measure conditional on time- t filtration $\mathcal{F}_t$.

Equation (2) prices a single option contract by requiring that the option's time decay be balanced by the risk-neutral expected profit and loss from four sources of risk exposures. The risk-neutral expected profit and loss from each risk source is given by the product the risk exposure and the risk-neutral expected magnitude of the corresponding source of variation.

2.2. Dimension reduction through risk-targeting option portfolios

For our purpose of dimension reduction, we pool the pricing of all the option contracts on a stock and reduce them to four option portfolios targeting each of the four risk dimensions. Let $n_{t,i}$ denote the number of selected option contracts on stock i at date t . We start with a set of conditional moment forecasts under the statistical measure $\mathbb{P}$

$$v_t^{\mathbb{P}} \equiv \frac{1}{dt}\mathbb{E}_t^{\mathbb{P}}\left[\left(\frac{dS_t}{S_t}\right)^2\right], \quad \omega_{t,j}^{\mathbb{P}} \equiv \frac{1}{dt}\mathbb{E}_t^{\mathbb{P}}\left[\left(\frac{dI_{t,j}}{I_{t,j}}\right)^2\right], \quad \gamma_{t,j}^{\mathbb{P}} \equiv \frac{1}{dt}\mathbb{E}_t^{\mathbb{P}}\left[\frac{dS_t}{S_t}, \frac{dI_{t,j}}{I_{t,j}}\right]. \quad (4)$$

Since both the option implied volatility level and variation can be different for different option contracts even underlying the same stock, the implied volatility variance ($\omega_{t,j}^{\mathbb{P}}$) and covariance ($\gamma_{t,j}^{\mathbb{P}}$) can vary across the different option contracts. Under deterministic rates and dividend assumptions, the stock return variance $v_t^{\mathbb{P}}$ is the same across all option maturities.

We link these statistical risk forecasts to the corresponding risk-neutral expectations with a set of common pricing assumptions across the $n_{t,i}$ option contracts as in Wu and Zhang (2024),

$$\mu_{t,j}^{\mathbb{P}} = 0 \quad (5)$$

$$\mu_{t,j} = \eta_{t,V} \omega_{t,j}^{\mathbb{P}}, \quad (6)$$

$$v_t = (\eta_{t,G} + 1) v_t^{\mathbb{P}}, \quad (7)$$

$$\omega_{t,j} = (\eta_{t,O} + 1) \omega_{t,j}^{\mathbb{P}}, \quad (8)$$

$$\gamma_{t,j} = (\eta_{t,A} + \rho_t^{\mathbb{P}}) \sqrt{\omega_{t,j}^{\mathbb{P}} v_t^{\mathbb{P}}}, \quad (9)$$

for all $j = 1, 2, \dots, n_{t,i}$. Equation (5) assumes no predictability on the implied volatility trend. Equation (6) imposes a common market price of vega risk $\eta_{t,V}$ across all the option contracts per unit of each contract's implied volatility variance $\omega_{t,j}^{\mathbb{P}}$. Equations (7) and (8) assume that the risk-neutral stock return variance and the implied volatility variance are proportional to their statistical counterpart, with their proportional deviations representing the market price of gamma risk ($\eta_{t,G}$) and volga risk ($\eta_{t,O}$), respectively. Equation (9) links the risk-neutral return-implied volatility covariance of each option contract to the square root of the product of the two statistical variance forecasts, where the proportionality coefficient is determined by the return-implied volatility correlation $\rho_t^{\mathbb{P}}$ and a common market price of vanna risk coefficient $\eta_{t,A}$ across all option contracts on stock i . The last equation amounts to represent the statistical covariance as $\gamma_{t,j}^{\mathbb{P}} = \rho_t^{\mathbb{P}} \sqrt{\omega_{t,j}^{\mathbb{P}} v_t^{\mathbb{P}}}$. In principle, the statistical correlation forecast $\rho_t^{\mathbb{P}}$ can also vary across option contracts, we impose a

common correlation coefficient to enhance identification.

While each option contract can be priced based on its own risk behavior based on the decentralized pricing equation in (2), the set of commonality assumptions and parameterizations we make in equation (5) to (9) allows us to pool the pricing of $n_{t,i}$ option contracts on the same stock i together and perform dimension reduction. Specifically, we can rewrite the pricing equations on the $n_{t,i}$ option contracts and factor out four common pricing coefficients,

$$-B_t^j = X_{t,j} \mathbf{b}_t, \quad j = 1, 2, \dots, n_{t,i}, \quad (10)$$

where $X_{t,j}$ denotes the four-dimensional risk-adjusted exposures of option contract j ,

$$X_{t,j} = \left[B_I^j I_{t,j} \omega_{t,j}^{\mathbb{P}}, \quad \frac{1}{2} B_{SS}^j S_t^2 v_t^{\mathbb{P}}, \quad \frac{1}{2} B_{II}^j I_{t,j}^2 \omega_{t,j}^{\mathbb{P}}, \quad B_{IS}^j I_{t,j} S_t \sqrt{v_t^{\mathbb{P}} \omega_{t,j}^{\mathbb{P}}} \right], \quad (11)$$

and $\mathbf{b}_t$ denotes the common pricing coefficients on the four risk dimensions across all $n_{t,i}$ option contracts,

$$\mathbf{b}_t = [\eta_{t,v}, \eta_{t,G} + 1, \eta_{t,O} + 1, \eta_{t,A} + \rho_t^{\mathbb{P}}]^\top. \quad (12)$$

The risk-adjusted exposures $X_{t,j}$ can vary across option contracts due to variations in both the BMS sensitivities and the implied volatility variance forecasts, but the loadings $\mathbf{b}_t$ are common across all the $n_{t,i}$ option contracts on the stock i . We can identify the common loading coefficients by regressing the $(n_{t,i} \times 1)$ vector of the options time decay $\mathbf{y}_t = -\{B_t^j\}_{j=1}^{n_{t,i}}$ against the $(n_{t,i} \times 4)$ risk-adjusted exposure matrix X_t ,

$$\mathbf{y}_t = X_t \mathbf{b}_t + \mathbf{e}_t. \quad (13)$$

The slope estimates from the regression, upon adjusting for their corresponding null values under the hypothesis of zero risk premium, captures the time- t common market price of risk on the four

risk dimensions for stock i ,

$$\boldsymbol{\eta}_t \equiv [\eta_{t,V}, \eta_{t,G}, \eta_{t,O}, \eta_{t,A}]^\top = \mathbf{b}_t - \mathbf{b}_t^0, \quad (14)$$

with the null values given by

$$\mathbf{b}_t^0 = \begin{bmatrix} 0 & 1 & 1 & \rho_t^{\mathbb{P}} \end{bmatrix}^\top. \quad (15)$$

The identification is achieved by pooling the $n_{t,i}$ pricing equations for the $n_{t,i}$ option contracts on stock i at date t assuming that they can have different risk exposures and different risk magnitudes, but share a set of common market pricing per unit risk along each of the four risk dimensions.

Importantly, the regression amounts to reduce the $n_{t,i}$ option contracts on stock i into four risk-targeting option portfolios, with the portfolio weight on each risk dimension given by

$$\mathbf{w}_{t,k} = X_t(X_t^\top X_t)^{-1} \mathbf{1}_k, \quad k = 1, 2, \dots, 4, \quad (16)$$

with $\mathbf{1}_k$ an indicator vector with the k th element being one and all other elements zero.

If we construct the ex post annualized realized delta-hedged excess return over the next time period from writing the $n_{t,i}$ option contracts, $\mathbf{r}_{t+1}$, we can compute the ex post delta-hedged excess return from writing the corresponding risk-targeting option portfolio as,

$$f_{t+1,k} = \mathbf{w}_{t,k}^\top \mathbf{r}_{t+1}. \quad (17)$$

Proposition 1 *At each date t and for each stock i , we consolidate information and summarize the ex ante pricing and ex post return behavior of its $n_{t,i}$ option contracts through four risk-targeting option portfolios. The market price of risk estimate $\eta_{t,k}$ on the k th risk dimension represents the*

ex ante market expectation of the ex post annualized excess return realization over the next period, $f_{t+1,k}$ on the corresponding risk-targeting option portfolio.

The online appendix provides the proof for the proposition.

At each date t , we form four risk-targeting option portfolios from the $n_{t,i}$ option contracts on each stock i in the selected universe of N_t stocks. The original options data structure is complex and difficult to work with for comparative analysis because each stock has different numbers of options, with different strikes and maturities, different BMS sensitivities, and different risk levels. Our operation reduces this complex data structure into a clean $(N_t \times 4)$ matrix of market pricing coefficients, targeting four distinct risk sources with normalized unit risk levels. We can similarly summarize the ex post option excess return behavior of the option contracts through an analogous $(N_t \times 4)$ matrix of excess returns on the corresponding risk-targeting option portfolios.

Both the market price of risk and the ex post realized excess returns on the risk-targeting portfolios can vary across the different stocks. The cross-sectional variations of the two sets of estimates are related through the expectation hypothesis:

Corollary 1 *If we regress the ex post annualized excess returns of the risk-targeting option portfolios $f_{t+1,k}$ against the ex ante market price of risk estimates $\eta_{t,k}$ across the stock universe,*

$$f_{t+1,k} = a_{t+1,k} + b_{t+1,k}\eta_{t,k} + \epsilon_{t+1,k}, \quad (18)$$

the null value of the regression intercept is zero and the null value of the slope is one under the expectation hypothesis.

Through the dimension reduction operation, we can examine how the market pricing of a given risk dimension varies cross-sectionally over different underlying stocks, how this variation is linked

to variations in the underlying firm risk characteristics, and whether the market price of risk estimates can predict the future realized excess returns of the corresponding risk-targeting option portfolio according to the expectation hypothesis.

3. Data processing and option portfolio construction

This paper analyzes the behavior of individual stock options listed in the U.S. stock options exchanges using data from OptionMetrics.

3.1. Sample selection

The options data are available from January 1996 to December 2021 at the time of our analysis. We leave the first year of data for historical moment estimation, and the last month of 2021 for option return calculation over the following month. We sample the data once a month, at 30 calendar days before the traditional option expiration date in the next month, i.e., the third Friday of the next month. When that date is a holiday, we choose the previous business day as the sampling date.

At each sampling date, we determine the stock universe based on the following criteria: (1) It is a common stock on a U.S. publicly traded company. (2) The stock has at least one year history of stock returns and options trading. (3) The stock has a minimum of 20 option contracts on the sampling date satisfying our options selection criteria. (4) The options span more than two maturities between one month and one year. We require a one-year history to compute the return variance and historical moment conditions on the option implied volatility changes. The last two criteria guarantee that we have sufficient number of option contracts over a range of maturities and strikes to construct the risk-targeting option portfolios.

For each stock in the universe, we select the option contracts based on the following criteria: (1) The option quote has valid implied volatility calculation from OptionMetrics. (2) The bid quote is strictly positive. (3) The ask quote is strictly greater than the bid quote. (4) The option's absolute forward delta is greater than 1%, but lower than 80%. (5) The option's days to maturity are no fewer than 30 days and no greater than 365 days. (6) The contract has strictly positive trading volume or open interest on the sampling date. (7) The option has valid price quote 30 days from the sampling date if it is a non-expiring contract and the underlying stock has valid price quote 30 days from now if the option expires in 30 days. The first three conditions set the minimum criteria for the validity of the pricing information. We filter out deep in-the-money options with absolute delta greater than 80% as they tend to be less liquid than the corresponding out-of-the-money options and they tend to be affected more by the early exercise premium of the American options. We also exclude deep out-of-the-money options with absolute delta below 1%, options with less than 30 days to maturity or greater than 365 days to maturity, and options with no trading volume or open interest. The last criterion guarantees that we can compute the option excess return over the next month. We adjust the delta by the dividend information to obtain the forward delta.

The above filtering criteria lead to a selection of 16,947,489 option contracts on 3,691 individual stocks over a period of 299 months (25 years) from January 1997 to November 2021. Figure 1 plots in panel A the number of selected stocks at each sampling date, which varies from 259 stocks in July 1997 to 1,794 stocks in October 2021, averaging at 954 selected names per sampling date.

The number of option contracts per stock varies greatly across different stocks and time periods. We set a minimum requirement of 20 selected contracts per name. The maximum number of selected option contracts reaches as high as 2,896 for Tesla on April 2020. Panel B of Figure 1 plots the distribution of the number of selected options per stock at each sampling date, including the mean (solid line) and the 10-90th percentile range (dash-dotted lines). Just as the number of

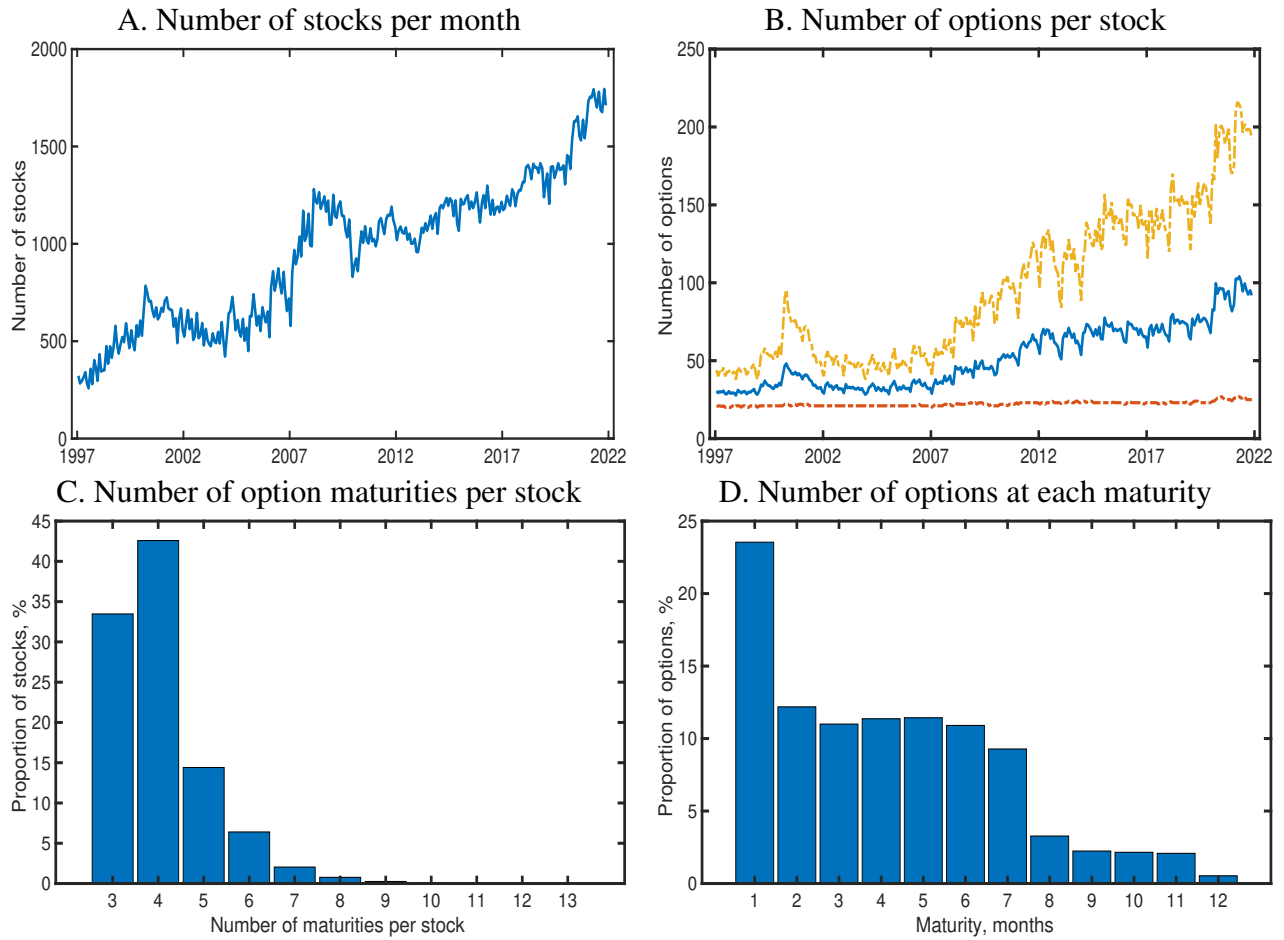


Figure 1

The number of stocks per sampling date and number of options per stock

Panel A plots the number of selected stock per sampling date. Panel B plots the distribution on the number of selected options per stock at each sampling date, including the median (solid line), the 25th, and the 75th percentiles (dash-dotted lines). Panel C plots the pooled distribution on the number of chosen maturities per stock. Panel D plots the number of chosen option contracts at each maturity from one to 12 months.

selected stocks increases over time, the average number of option contracts per stock also increases over time from a low of 28 contracts to a high of 104 contracts.

Panel C plots the pooled distribution on the number of option maturities for each stock. The number of maturities usually varies from three to five maturities between 30 and 365 days. A small number of names have options spanning as many as 13 maturities. Panel D counts the number of options at each maturity bucket from one to 12 months and plots the pooled distribution. One-month options represent 24% of the sample. Two to seven month buckets each contribute 9-12% for a total of 66%. The remaining 10% of the sample has maturities from eight to 12 months.

3.2. Constructing risk-targeting option portfolios

At each date t , for each selected option contract j on a stock i , we construct its risk-adjusted exposures $X_{t,j}$ as in (11). The risk-adjusted exposures are determined by the contract's BMS sensitivities: cash vega ($B_I^j I_{t,j}$), cash gamma ($B_{SS}^j S_t^2$), cash volga ($B_{II}^j I_{t,j}^2$), and cash vanna ($B_{IS}^j S_t I_{t,j}$), as well as the contract's two variance forecasts ($\omega_{t,j}^{\mathbb{P}}, v_t^{\mathbb{P}}$).

The individual stock options are American style. The data vendor OptionMetrics uses a tree approach to price the American options and compute the option implied volatility. To simplify the computation, we take the option implied volatility from OptionMetrics as our starting point, and compute the forward risk sensitivities on the “de-Americanized” European counterparts, which are all in simple analytical forms and provided in the online appendix. In computing the BMS sensitivities, we normalize the forward security price to one ($S_t = 1$) and convert the sensitivities from dollar terms to excess returns on forward notional.

We proxy the two conditional variance forecasts ($\omega_{t,j}^{\mathbb{P}}, v_t^{\mathbb{P}}$) with their corresponding one-year historical moment estimators ($\hat{\omega}_{t,j}, \hat{v}_t$). Constructing the one-year historical variance estimator

on daily stock returns is straightforward. Constructing historical variance estimators on the proportional implied volatility changes is more involved. The listed option contracts have fixed strike prices and expiration dates. As time goes by, the moneyness and maturity of the contracts vary. We therefore need some pooling and interpolation to generate historical implied volatility changes for the variance estimator at a target moneyness and maturity. The online appendix provides a general procedure for the historical estimator construction.

With the $(n_{t,i} \times 4)$ risk-adjusted exposure matrix X_t constructed on the $n_{t,i}$ option contracts on a stock i , we construct the weights for the four risk-targeting option portfolios on the stock as in (16). To gain a structural understanding of the risk-targeting option portfolios, we pool the portfolio weights across all stocks and all dates and compute the pooled average portfolio weights across a fixed grid of option maturities and moneyness. We normalize the average portfolio weights so that the absolute weights across the chosen fixed maturity-moneyness grids sum to one. The absolute scale of the weights on an option contract depends not only on contract's risk exposures but also the number of selected option contracts for that underlying stock. The normalization removes the absolute scale but highlights the relative loading pattern. Figure 2 plots the normalized average portfolio weights. Each panel is for one portfolio. The lines plot the relative weight variation across the put forward delta at the selected maturities of one month (solid line), three months (dashed line), six months (dash-dotted line), and 12 months (dotted line).

The vega portfolio in panel A is long long-dated options and short short-dated options, with the absolute weights higher for at-the-money options than for out-of-the-money options. The portfolio is similar to a classic at-the-money calendar spread portfolio. The gamma portfolio in panel B is also centered around at-the-money options, but is in the opposite direction of the vega portfolio by being long short-dated options and short long-dated options. As options have higher vega exposures at longer maturities but higher gamma exposures at shorter maturities, the calendar

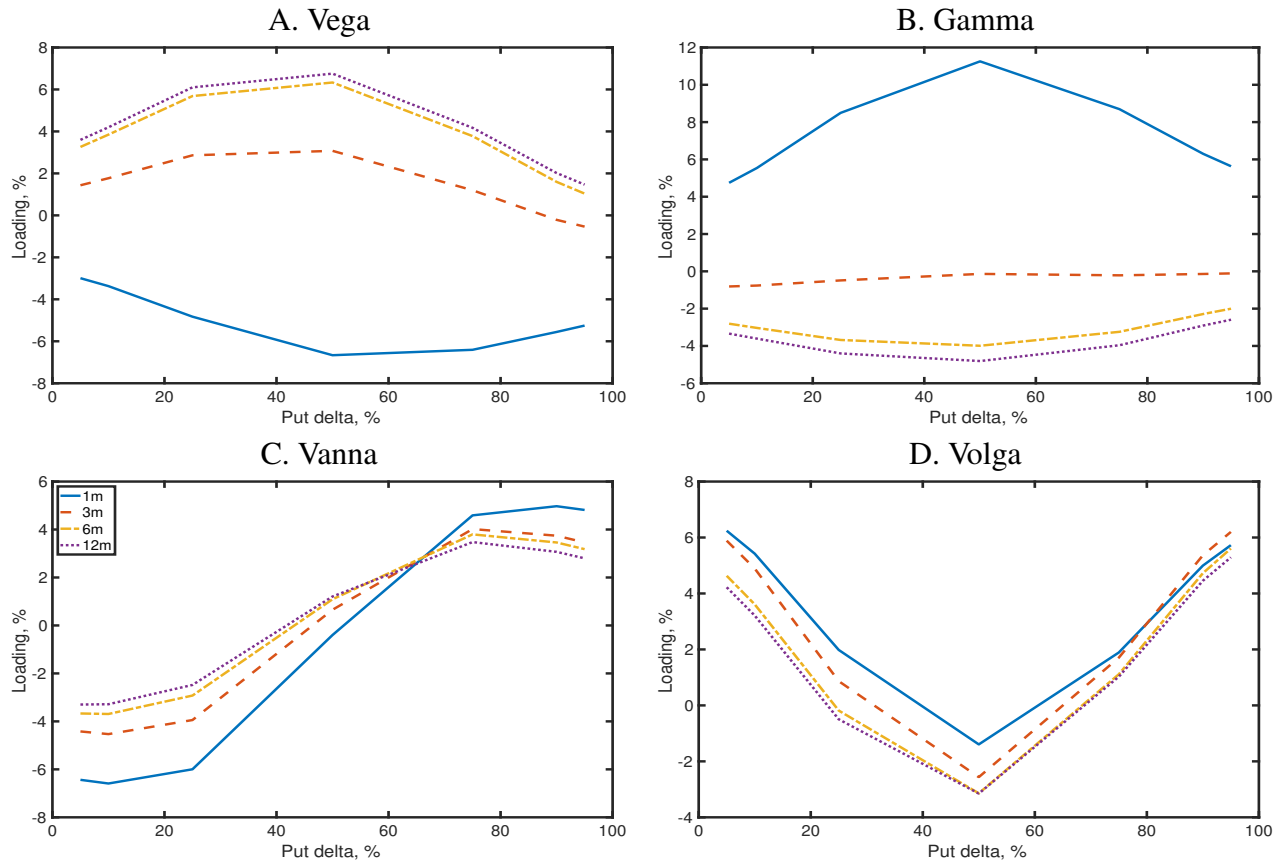


Figure 2

The average loading patterns of the risk-targeting option portfolios

Each panel plots the average relative weights on a risk-targeting option portfolio. Each line plots the weight variation across option moneyness represented in absolute put forward delta at one maturity: one month (solid line), three months (dashed line), six months (dash-dotted line), and 12 months (dotted line).

spread that is long long-dated options in panel A targets a positive vega exposure whereas the spread that is long short-dated options in panel B targets a positive gamma exposure.

The vanna portfolio in panel C is long out-of-the-money call options and short out-of-the-money put options, similar to a classic risk reversal construction. The volga portfolio in panel D looks very much like a classic butterfly construction that is long deep out-of-the-money options and short at-the-money options.

Calendar spreads, risk reversals, and butterfly spreads have been standard combinations in the over-the-counter currency options market that is catered to institutional investors. Our theoretical construct shows that these combinations are natural choices for targeting a specific risk dimension while being (largely) neutral to other risk dimensions.

3.3. Estimating the market pricing and excess return of the risk portfolios

At each date t and for each stock i in our universe, we perform a cross-sectional contemporaneous pricing regression on the $n_{t,i}$ option contracts, where we regress the option time decay vector $\mathbf{y}_t = -B_t$ against the risk-adjusted exposure matrix X_t , as in (13),

$$\mathbf{y}_t = X_t \mathbf{b}_t + \mathbf{e}_t. \quad (19)$$

We construct the market prices of risk on the four risk dimensions as the difference between the slope coefficient estimates from the pricing regression and their corresponding null values,

$$\hat{\eta}_t = \hat{\mathbf{b}}_t - \mathbf{b}_t^0, \quad \text{with} \quad \mathbf{b}_t^0 = \begin{bmatrix} 0 & 1 & 1 & \hat{\rho}_t \end{bmatrix}^\top. \quad (20)$$

To construct the market price of vanna risk, we take the historical variance and covariance estimators at 50-delta and three month maturity to generate a correlation estimator between the stock return and option implied volatility change,

$$\widehat{\rho}_t = \widehat{\gamma}_t / \sqrt{\widehat{\sigma}_t \widehat{\omega}_t}. \quad (21)$$

In parallel to the contemporaneous pricing regression in (19), we also perform a cross-sectional forecasting return regression to explain the cross-sectional variation of the delta-hedged option excess returns over the next month $\mathbf{r}_{t+1}$ with the same set of risk-adjusted risk exposures X_t and the residual $\mathbf{e}_t$ from the pricing regression in (19),

$$\mathbf{r}_{t+1} = X_t \mathbf{f}_{t+1} + \mathbf{e}_t \zeta_{t+1} + \boldsymbol{\varepsilon}_{t+1}, \quad (22)$$

where the delta-hedged excess return over the next month from writing the j th option contract is constructed as,

$$r_{t+1,j} = \frac{12}{S_t} \left[\left(O_t^j - O_T^j \right) + \sum_{d=1}^n \Delta_{t_d}^j (S_{t_{d+1}} - S_{t_d}) \right], \quad (23)$$

where (O_t^j, O_T^j) denote the forward values of the option contract at the starting and ending date of the investment. Their difference captures the dollar excess profit and loss from the option position. The second term captures the contribution from daily delta rebalancing, with $\{t_d\}_{d=1}^n$ denoting the list of rebalancing days within the monthly investment horizon, $\Delta_{t_d}^j$ denoting the forward delta, and $(S_{t_{d+1}} - S_{t_d})$ denoting the forward stock price changes within each delta rebalancing interval. We multiply the monthly excess profit and loss by 12 and scale it by the forward stock price to construct the annualized excess return on forward notional. We use the option mid quote and adjust for the interest rate to obtain the forward option value. We adjust the stock closing price with the

dividend and interest rate to obtain the forward stock value corresponding to each option contract. We update the delta position on the underlying stock whenever valid stock and option pricing information is available. As the option writer bears the bulk of the investment risk, we examine the option investment risk and return from the perspective of an option writer and constructs the option excess return in (23) from writing the option contract.

From the return factor model in (22), the slope coefficients $\mathbf{f}_{t+1}$ on the risk-adjusted exposure matrix capture the annualized excess returns over the next month on the risk-targeting option portfolios that target a unit exposure to each risk source and zero exposures to other sources. The model also includes the residual $\mathbf{e}_t$ from the pricing model as a mispricing factor. The slope ζ_{t+1} on the mispricing factor captures the annualized excess return on a statistical arbitrage value-investing portfolio that targets a unit exposure to the option pricing error while neutral to all four risk sources. The portfolio is constructed to write options with positive pricing errors and buy options with negative pricing errors while being neutral to the risk-adjusted exposures.

3.4. Practical implementation and investment considerations

Constructing the risk-targeting option portfolios requires estimating the risk-adjusted exposure matrix X_t , which needs inputs on both the BMS sensitivities and the statistical forecasts of the stock return variance and the implied volatility change variance. Estimating the market price of risk further requires a forecast on the return-implied volatility correlation. To estimate the BMS risk sensitivities, we start with the implied volatility that OptionMetrics provides for each option contract and computes the sensitivities of the corresponding European option counterpart. OptionMetrics estimates the option implied volatility with a tree approach that accounts for the early exercise of the American option. One can in principle estimate the sensitivities on the same tree.

As we do not have access to their pricing engine, we use the analytical sensitivities of the European counterpart as a shortcut. This “de-Americanization” procedure is a common maneuver in estimating option pricing models to reduce the computational burden. We do not expect this procedure to induce large bias on the market price of risk estimates as the pricing regression uses the European option sensitivities on both sides of the equation.

We construct the option excess returns on the actual American option mid quotes. The early exercise premium of the American options may induce a small bias between the excess returns on the risk-targeting option portfolios and the corresponding market price of risk estimates.

The accuracy of the statistical moment forecasts can also impact the accuracy of the risk-targeting option portfolio construction and the market price of risk estimates. This paper adopts the simple, conservative one-year historical moment estimator to proxy the statistical forecast. One-year historical moment estimators are common choices for asset pricing and investment analysis (e.g., Goyal and Saretto (2009) and Bai and Wu (2016)) and for benchmarking variance forecasting accuracies (e.g., Wu and Xu (2022)). There is a large literature on variance forecasts. The results we obtain from the simple historical moment estimators can be regarded as the benchmark that researchers can improve upon with enhanced moment forecasts.

Finally, we apply a minimum set of filters in selecting the stock universe and the option contracts per stock. This allows us to analyze the applicability of the theoretical construct over a large stock universe with different number of option contracts, different strike and maturity ranges, and different trading liquidity and pricing accuracy. The analysis highlights the general applicability and the universal effectiveness of the dimension reduction operation through the portfolio construction, and the strong information content in the market price of risk estimates.

For practical investment implementations, investors may start with a much smaller set of stocks

with large option trading volumes. For each selected stock, to reduce trading cost, investors may choose to construct the risk-targeting option portfolios on a much more selected group of option contracts. Indeed, investors in the stock options market can learn from the over-the-counter currency options market and focus on a set of extremely simplified option combinations to trade on the four risk dimensions, e.g., calendar spreads around at-the-money strikes, risk reversals and butterflies constructed around 10% or 25% delta. The options exchanges nowadays also list option combinations that can facilitate such risk-targeting investments. Since these risk-targeting combinations tend to have smaller risk exposures than a naked vanilla option contract, market makers can afford to offer narrower spreads on well-constructed risk-targeting option combinations than on a single option contract. To estimate the market price of risk, investors can still select a wide range of option contracts to enhance identification, potentially with a weighted regression by applying different weights to option contracts with different pricing accuracy. What we offer is a general framework. Much room is left for future research and enhancement in practical implementation.

4. Cross-sectional variation of risk-targeting option portfolios

For the risk-targeting option portfolios to adequately summarize the variations of the individual options contracts, it is imperative that the underlying risk structure can explain the options pricing behavior and return variations well. This requires the option pricing model in (19) to have high explanatory power on the options pricing behavior, and the return factor model in (22) to explain a large proportion of the option excess return variation. This section starts by summarizing the estimation results on the option pricing model, including the model's pricing performance and the summary behaviors of the market price of risk estimates. The section then summarizes the estimation results on the return factor model, including the model's explanatory power on the

excess return variation and the realized excess return behaviors of the risk-targeting portfolios. The section ends by examining the expectation hypothesis that the market price of risk represents the conditional expectation on the ex post excess return of the corresponding risk-targeting portfolio.

4.1. The market price of risk behavior

At each date t , we perform the pricing regression (19) for each name i in the selected stock universe of N_t stocks. The regression R^2 estimates capture how well the four risk-targeting option portfolios can summarize the pricing behavior of the different option contracts on the stock. The slope coefficients reflect the market price of risk on the stock along the four risk dimensions. Table 1 summarizes the average behaviors of the market price of risk and regression R^2 estimates. Panel A reports the time-series statistics of the market price of risk estimates on the equal-weighted market portfolio across the stock universe for each risk dimension. The statistics include the time-series average (Mean), standard deviation (SD), and the Newey and West (1987) t -statistics on the average (NW). In constructing the t -statistics, we adjust for the serial dependence of the time series with a lag of three months.

The average market price of gamma and volga risk estimates are positive and statistically significant. The average market pricing estimate of the vanna portfolio is strongly negative and highly statistically significant. The average market price of vega risk is close to zero.

Under our theoretical construct, a positive market price of risk suggests that the portfolio is priced more expensively than the breakeven benchmark. The equal-weighted average market price of risk estimates suggest that the gamma and volga equal-weighted market portfolios are on average more expensively priced than breakeven whereas the vanna portfolio is on average priced lower than than breakeven. As such, we expect to make positive average risk premiums by writing the

Table 1
The market price of risk estimates and the pricing performance

$\eta_{t,k}$	A. Equal-weighted portfolio			B. Cross-sectional percentiles		
	Mean	SD	NW	10	50	90
Vega	-0.00	1.05	-0.03	-1.89	0.05	1.82
Gamma	0.10	0.45	2.45	-0.45	-0.00	0.74
Volga	0.13	0.63	1.92	-0.84	-0.06	1.37
Vanna	-0.23	0.17	-13.05	-0.52	-0.21	0.06
R^2	0.985	0.009		0.967	0.992	0.998

Panel A reports the time-series statistics of the market price of risk estimates on the equal-weighted market portfolio across the stock universe for each risk dimension. The statistics include the mean (Mean), standard deviation (SD), and the Newey-West t -statistics on the average (NW). Panel B reports the time-series average of the cross-sectional percentile values of the market price of risk estimates at the 10, 50, and 90th percentiles. The last row reports the statistics on the R^2 estimates of the pricing regression.

expensively priced gamma and volga portfolios, and by buying the vanna portfolios which are priced at a discount on average. Combining these average estimates with the portfolio weights in Figure 2, we can further infer that the investors on average expect to make positive risk premiums by shorting short-dated at-the-money options, shorting butterflies, and long risk reversals.

At each date, we also compute the cross-sectional percentile values of the market price of risk estimates at the 10, 50, and 90th percentiles. Panel B reports the time-series averages of the percentile values. Although the average market price of vega risk is close to zero, the estimates show large cross-sectional variations. The average percentile values vary from -1.89 to 1.82 over the 10-90th percentile range. The market price of gamma risk estimates have an average median value close to zero and vary from an average of -0.45 to an average of 0.74 over the 10-90th percentile range. The market price of volga risk estimates also have a small average median value at -0.057 , but vary from -0.84 to 1.37 over the 10-90th percentile range. For all three risk sources, the average or median values of the market price of risk estimates are small, relative to the wide

cross-sectional variation.

The market price of vanna risk estimates are mostly negative. Both the average median value and equal-weighted average are highly negative, and the average value at the 90th percentile is only slightly positive. The highly negative market pricing of vanna risk suggests that the historical return-implied volatility correlation estimators strongly under-predict the negative slope of the implied volatility skew.

The last row reports the summary statistics on the R^2 estimates of the pricing regression. The four risk-targeting option portfolios summarize the option pricing behavior very well. The average R^2 reaches as high as 98.5%, and the average median is even higher at 99.2%. The equal-weighted average R^2 estimates show little time variation with a standard deviation estimate of merely 0.9%. The cross-sectional variation is also very small, from an average of 96.7% at the 10th percentile to an average of 99.8% at the 90th percentile. Therefore, there is little information loss by reducing the option contracts into four risk-targeting option portfolios. This is true across the whole selected stock universe despite the large variation in the number of options contracts per each name.

4.2. The excess return behavior

At each date and for each stock, we also perform a cross-sectional forecasting return regression on the return factor model in (22). The regression R^2 captures how well the proposed risk structure, together with the residual from the pricing model, can explain the cross-sectional option return variation over the next month. The slope coefficients from the return regression represents the excess returns from writing the corresponding risk-targeting option portfolios.

Table 2 summarizes the excess return behavior of the option portfolios and the explanatory power of the return factor model. Panel A reports the summary excess return behavior of the

Table 2**Excess return behavior of the risk-targeting option portfolios and the explanatory power**

$f_{t+1,k}$	A. EW portfolio			B. CS percentiles		
	Mean	Vol	NW	10	50	90
Vega	-0.97	1.60	-2.62	-8.31	-0.38	5.75
Gamma	0.08	0.16	2.57	-0.93	0.13	1.13
Volga	0.18	0.41	1.93	-3.52	0.13	4.09
Vanna	0.01	0.13	0.21	-1.00	-0.01	0.99
Error	1.25	0.13	37.73	-0.54	1.16	3.23
R^2	0.785	0.055		0.529	0.837	0.957

Panel A reports the summary statistics of the excess return time series from writing the equal-weighted portfolio of each risk-targeting option portfolios, including the annualized mean excess return (Mean), annualized return volatility (Vol), and the Newey-West t -statistics on the mean excess return (NW). Panel B reports the time-series averages of the cross-sectional (CS) percentile values of the annualized excess return.

equal-weighted market portfolio of each of the risk-targeting option portfolios. The statistics include the annualized mean excess return (Mean), the annualized return volatility (Vol), and the Newey and West (1987) t -statistics on the mean excess return (NW). The mean excess return on the vega portfolio is significantly negative, whereas the mean excess returns on the gamma and volga portfolios are significantly positive. The average excess return on the vanna portfolio is close to zero.

Different from the four risk-targeting portfolios, the statistical arbitrage portfolio that targets the pricing error from the pricing model generates the the highest average excess return with low return volatility, leading to extremely high t -value estimate. The portfolio does not take on any systematic risk exposures on any stocks, but rather bets on the convergence of the pricing errors. The extremely high t -value suggests that the pricing errors from the option pricing model are strongly mean reverting on average. In particular, as the pricing regression is performed in the expected excess return space, the average slope estimate of 1.25 suggests that each one percentage

point mispricing in the expected excess return space translates into 1.25 percentage point of ex post annualized excess return. The small pricing residual unexplained by the four risk-targeting option portfolios can be treated as highly reverting market mispricing and highly profitable statistical arbitrage trading opportunities.

Panel B reports the time-series averages of the cross-sectional percentile values of the excess returns. For each of the four risk-targeting option portfolios, the ex post realized excess returns vary over an ever wider range than the corresponding market price of risk estimates. The average median excess returns are all small relative to the 10-90th percentile range. Along each risk dimension, the average excess returns on the risk-targeting option portfolio can vary from highly negative to highly positive across the stock universe. Therefore, being long or short a risk dimension indiscriminately across the stock universe can only capture the small average risk premium, but misses the much larger opportunity in the cross-sectional variation.

The excess returns on the statistical arbitrage portfolios are predominantly positive across the stock universe. The average median excess return is highly positive. The average excess return at the 10th percentile becomes negative, but the value is small relative to the average median value. Therefore, different from the four risk-targeting option portfolios, one can in general profit from investing in the statistical arbitrage portfolio indiscriminately across the whole stock universe.

The last row reports the summary statistics on the explanatory power of the return factor model. In aggregate, the return factor model can explain a large proportion of the option excess return variation. The gross R^2 estimates on the individual stock options average at 78.5% and vary from 52.9% at the 10th percentile to 95.7% at the 90th percentile, with an average median value of 83.7%. Therefore, the four risk-targeting option portfolios not only summarize the option pricing behavior extremely well, but they also explain a large proportion of the option excess return

variation over the next month.

4.3. The expectation hypothesis

Proposition 1 shows that under our theoretical construct, the market price of risk represents the conditional market expectation of the future realized excess return from writing the corresponding risk-targeting option portfolio. Under the expectation hypothesis, if we regress the future realized excess return of each risk-targeting portfolio against its ex ante market price of risk estimate, we would expect a null value of zero for the intercept and a null value of one for the slope.

To test this expectation hypothesis, at each date t , for each risk-targeting option portfolio k , we perform the following cross-sectional forecasting return regression,

$$\mathbf{f}_{t+1,k} = a_{t+1,k} + b_{t+1,k}\boldsymbol{\eta}_{t,k} + \boldsymbol{\varepsilon}_{t+1,k}, \quad (24)$$

where $\mathbf{f}_{t+1,k}$ denotes the $(N_t \times 1)$ vector of annualized excess returns over the next month on the k th risk-targeting option portfolio across the universe of N_t stocks at time t , and $\boldsymbol{\eta}_{t,k}$ denotes the corresponding vector of market price of risk estimates. For each risk dimension k , the regression examines the expectation hypothesis over the cross section of the stock universe at each date.

Table 3 reports the time-series averages and the Newey and West (1987) t -statistics (in parentheses) of the regression intercept, the slope, and the R^2 estimates. Consistent with the broad implications of the expectation hypothesis, the average slope estimates on the four sets of risk-targeting option portfolios are all positive and strongly statistically significant. The ex ante market price of risk estimates on a risk source strongly and positively predict the future realized excess returns on the corresponding risk-targeting option portfolio.

Table 3
The expectation hypothesis regression

Risk	a		b		R^2 , %
Vega	-0.772	(-2.66)	0.867	(15.62)	8.02
Gamma	0.017	(0.62)	0.396	(9.50)	11.00
Volga	0.068	(0.84)	0.769	(15.41)	6.16
Vanna	0.111	(3.76)	0.488	(11.57)	2.93

At each date and for each risk dimension, we regress the cross section of the excess returns on the risk-targeting option portfolios over the next month against the corresponding market price of risk estimates. Entries report the time-series averages and the Newey-West t -statistics (in parentheses) of the intercept (a), the slope (b), and the R^2 estimates.

The average slope and intercept estimates also show some deviations from the null values. First, the average slope estimates are all below the null value of one. This downward bias can partially be driven by the estimation errors in the market price of risk estimates and the effect of errors-in-variables problem. The market price of risk estimates $\eta_{t,k}$ are obtained from the slope estimates of the cross-sectional regression estimation of the pricing relation, and are naturally estimated with errors. In experimentation, we find that a simple cross-sectional winsorization on the market price of risk estimates can reduce the impacts of extreme estimates, and move the average slope estimates closer to the null value of one. With a cross-sectional winsorization at the 1-99% range, the average slope estimates on the gamma portfolio increases to 0.485 and that on the vega portfolio increases to 0.998, no longer significantly different from the null value of one.

The average intercept estimates are not statistically different from the null value of zero for the gamma and volga portfolios, but the average intercept estimate on the vega portfolio is significantly negative whereas the average intercept estimate on the vanna portfolio is significantly positive. For the vega portfolio, the average excess returns tend to be more negative than the average market price of risk estimates, leading to a significantly negative average intercept estimate. Part of the

average bias can come from our no-predictability assumption on the implied volatility trend. When the market anticipates a non-zero implied volatility trend, our market price of vega risk estimate captures the combined effect of this trend anticipation and the risk premium.

For the vanna portfolio, the market price of risk estimates can be downward biased if the historical return-volatility correlation estimator is less negative than it should be due to noises in the implied volatility change construction. More fundamentally, a potential source of bias on the market price of vanna risk estimate can come from the random arrival of default for companies with high financial leverage. Such an event can lead to a sudden drop in the stock price (Merton (1976)). The anticipation of such an event adds to the negative slope of the implied volatility skew (Tian and Wu (2024)). The BMS representation does not fully account for this source of risk and can mis-attribute the default-induced implied volatility skew to a more negative market pricing of vanna risk. Meanwhile, the average realized excess returns on the option portfolios also fail to capture the effects of these rare events by virtue of the peso problem and survival bias. As a result, the average realized excess return on the vanna portfolio is slightly positive even though the average market pricing estimate of vanna risk is strongly negative, leading to the positive average bias on the intercept estimate.

4.4. Long-short portfolios of risk-targeting option portfolios

Along each risk dimension, the market price of risk estimates vary widely across the stock universe at each date. As the market price of risk represents the market expectation of the ex post realized excess return on the corresponding risk-targeting option portfolio, we can use the market price of risk estimates to form a long-short portfolio on the risk-targeting option portfolios along each risk dimension to benefit from its positive predictive power and the large cross-sectional variation.

In principle, one can determine the predicted excess return based on some rolling historical averages of the cross-sectional forecasting regression in (24); nevertheless, the estimation errors of these predictive regressions can lead to performance deterioration in an out-of-sample setting. We propose two simple long-short portfolio construction strategies based on the market price of risk estimates without relying on any predictive regressions. The investment is naturally out of sample without any issue of in-sample overfitting or out-of-sample deterioration.

For each risk dimension k , at each sampling date t , we first re-scale the portfolio weights of each risk-targeting option portfolio to have one dollar notional. The rescaling applies to both the market price of risk estimate and the ex post excess return, so that when everything else equal, we can apply equal dollar investment to risk-targeting option portfolios across different stocks.

For the first strategy, we set the notional allocation weights to the N_t risk-targeting option portfolios simply in proportional to the corresponding re-scaled market price of risk estimates,

$$w_{t,k,i} = c_{k,t} \tilde{\eta}_{t,k,i}, \quad c_{k,t} = \frac{1}{\sum_{i=1}^{N_t} |\tilde{\eta}_{t,k,i}|}, \quad (25)$$

where $\tilde{\eta}_{t,k,i}$ denotes the time- t market price of risk estimate per dollar notional on the k risk dimension on stock i , $w_{t,k,i}$ denotes the notional allocation weight to each stock i on the portfolio of options, and the proportionality coefficient $c_{k,t}$ acts as a normalization so that the absolute investment weights to the portfolio of portfolios sum to one dollar. As the time- t market price of risk estimates can vary across the stock universe from negative to positive values, the portfolio weights can also vary from negative to positive, leading to a long-short portfolio of risk-targeting option portfolios along the k th risk dimension.

When the cross-sectional average of the market pricing estimates for a certain risk dimension at a given date deviates significantly from zero, the portfolio constructed on that date according to the

first strategy in (25) will have a significant aggregate exposure to the underlying risk source. In the second strategy, we remove this aggregate exposure and make the long-short portfolio aggregate notional neutral. We do so by first removing the cross-sectional mean of the market price of risk estimates and then making the notional allocation weights proportional to the demeaned market price of risk estimates,

$$w_{t,k,i} = c_{k,t} (\tilde{\eta}_{t,k,i} - \bar{\eta}_{t,k}), \quad c_{k,t} = \frac{1}{\sum_{i=1}^{N_t} |\tilde{\eta}_{t,k,i} - \bar{\eta}_{t,k}|}, \quad \bar{\eta}_{t,k} = \frac{1}{N_t} \sum_{i=1}^{N_t} \tilde{\eta}_{t,k,i}. \quad (26)$$

By removing the cross-sectional mean of the market price of risk estimates, the portfolio weights sum to zero and hence have zero aggregate dollar exposure to the risk dimension. The proportionality coefficient performs the same notional normalization so that the aggregate absolute notional investment is one dollar.

We use the two investment strategies to examine the investment implications of the expectation hypothesis on the market price of risk estimates, with the first strategy examining the hypothesis in its absolute form and the second strategy examining the hypothesis in the relative form. Table 4 reports the investment performance of the long-short portfolios of the risk-targeting option portfolios along each risk dimension. The statistics include the annualized mean excess return (Mean), the annualized return volatility (Vol), the annualized information ratio (IR), and the Newey and West (1987) t -statistics on the mean excess return (NW). Panel A reports the statistics on the first strategy. Panel B reports the statistics on the second strategy that removes the aggregate exposure.

Investing in an equal-weighted portfolio of the risk-targeting option portfolios can have either negative or positive average excess returns, with or without statistical significance, depending on the particular risk dimension (Panel A of Table 2). By comparison, Table 4 shows that the long-short investment in the risk-targeting portfolios is proportional to the market pricing esti-

Table 4
Long-short portfolios of risk-targeting option portfolios

Risk Dimension	A. With aggregate exposure				B. Without aggregate exposure			
	Mean	Vol	IR	NW	Mean	Vol	IR	NW
Vega	4.80	1.19	4.05	11.52	4.61	0.88	5.26	10.78
Gamma	1.35	0.62	2.18	10.18	1.55	0.45	3.46	14.12
Volga	1.71	0.74	2.32	4.89	1.62	0.57	2.84	5.68
Vanna	1.08	0.51	2.14	10.29	1.41	0.45	3.10	13.15

We construct long-short portfolios of the risk-targeting option portfolios in proportion to the market price of risk estimates. Entries report the summary statistics on the portfolio excess returns, including the mean excess return (Mean), the return volatility (Vol), the annualized information ratio (IR), and the Newey-West t statistics on the mean return (NW). Panel A retains the aggregate exposure to the risk dimension. Panel B removes aggregate exposure by deducting the cross-sectional average from the market price estimates in determining the long-short portfolio weights.

mates generates highly positive mean excess returns, with strong statistical significance and high information ratios, across all risk dimensions.

Comparing the investment results from the two panels shows that removing the aggregate dollar exposure in the investment does not change much of the mean excess return per dollar notional. The mean excess returns decline slightly from 4.8% to 4.61% for the vega portfolio and from 1.71% to 1.62% for the volga portfolio, but become higher for the vega and vanna portfolios. On the other hand, removing the aggregate dollar exposure significantly reduces the return volatility for all the four portfolios. As a result, the annualized information ratio estimates in panel B are all higher than the corresponding estimates in panel A.

Among the four risk dimensions, the long-short portfolio of the vega portfolios has the highest return volatility, reflecting the fact that the implied volatility movement through the vega exposure represents the largest source of return variation on the options. Luckily, the market pricing estimates on the vega portfolio can be used to predict this largest source of return variation, despite the

highly simplifying assumption of no trend predictability on the implied volatility movement. The long-short portfolios of the vega portfolio also generate the highest mean excess return per dollar notional and the highest information ratio.

The portfolio construction and investment exercise in this section is extremely stylized, without any consideration on the large transaction costs in the stock options market. The objective of the exercise is to highlight the strength of the option pricing model in identifying the ex ante market price of each risk source as a market expectation of the ex post realized excess return on the corresponding risk-targeting option portfolio.

The large cross-sectional variation of the market price of risk estimates and their prediction of the ex post excess return on the risk-targeting option portfolio represent a completely new dimension absent from the existing literature. When researchers propose investment strategies based on empirical return factor models, the strategy necessarily relies on historical rolling estimation of return prediction regressions. The general strategy is to be long a factor portfolio if the rolling historical average of the excess returns on the portfolio is significantly positive, and short if the historical average excess return is significantly negative. Such a strategy breaks down when the risk premium of a risk dimension varies strongly over time and switches signs, or when the aggregate risk premium is small relative to its large cross-sectional variation. When the risk premium varies strongly over time, historical rolling risk premium estimate can no longer effectively predict the future, and the history-based investment strategy is bound to fail out of sample. When the aggregate risk premium is small, the rolling estimation would not even be able to identify an investment opportunity. By comparison, with our market price of risk estimates, investors can perform market timing investment in any of the four risk dimensions in aggregate. They can also directly benefit from the large cross-sectional market pricing variation across the stock universe at each date without taking on any aggregate exposure.

5. The structural sources of market pricing variation

To identify the market price of each risk source on each stock at a given date, we impose common market pricing on the risk across all the option contracts on that stock, but without any assumptions on whether and how the market price should vary across different stocks. The observed cross-sectional variations of the market price estimates can be driven by either structural differences in the risk profiles of the underlying companies, or variations in temporary supply-demand shocks along a risk dimension. This section examines the implications of classic capital asset pricing theories on the cross-sectional structural variation of the market price of risk estimates.

5.1. Systematic versus idiosyncratic stock return risk

The classic capital asset pricing model decomposes the return risk of a stock into a systematic component and an idiosyncratic component and argues that the idiosyncratic return risk is not priced because it can be diversified away without cost, under the assumption of frictionless trading in the stock market. By contrast, starting with the BMS model, the dynamic no-arbitrage pricing of an option contract is not based on the risk diversification, but based on a complete replication and hedging of the option investment risk with the underlying stock. The option is valued based on the total stock return risk, without differentiating how much of the total return risk comes from systematic or idiosyncratic stock return variation.

Nevertheless, even under this dynamic no-arbitrage trading setting, when the stock return variance is not a constant but can vary stochastically, the systematic and idiosyncratic return components can contribute differently to the return variance dynamics. The different variance dynamics can lead to different market pricing contribution. In particular, the well-documented volatility feed-

back effect and the resulting negative return-volatility correlation mainly operate on the systematic return component.³ According to classic asset pricing theory, e.g., Merton (1973a), the market return risk premium is proportional to the market portfolio variance. As the variance increases, the stock market valuation declines due to increase in the discount rate, generating a negative correlation between the stock market return and the return variance. This effect, however, works only on the variance driven by the systematic return component because the idiosyncratic return variance does not raise the discount rate if the idiosyncratic return risk is not priced. As a result, stocks with a larger proportion of systematic return variation tend to have a more negative return-volatility correlation and a more negative option implied volatility skew (Tian and Wu (2024)).

This return-volatility correlation on the systematic return risk component can also propagate the return risk premium into a volatility risk premium. If the systematic return risk is priced positively, this positive return risk premium will be translated into a negative volatility risk premium through the negative correlation between the volatility shock and the systematic return shock. This volatility risk premium contribution from the systematic return risk is naturally larger for stocks with a larger proportion of systematic return variation.

To quantify how the proportion of systematic return risk affects the market price of risk estimates, for each stock and at each date, we estimate the correlation between the daily stock return and the daily return on the value-weighted market stock portfolio with a one-year rolling window. Then, we perform cross-sectional regressions of the market price of risk estimates on each risk dimension against this correlation estimate.

In empirical asset pricing tests, researchers often use the stock return beta to capture both the return correlation with the market portfolio and the risk level of the stock return. In our application,

³See Campbell and Hentschel (1992), Bekaert and Wu (2000), Carr and Wu (2017) for empirical documentations and economic explanations.

Table 5
Market price of risk variation with systematic stock return risk

Risk	Intercept		Slope		R^2 , %
Vega	-0.13	(-1.52)	0.36	(1.84)	1.43
Gamma	0.04	(1.54)	0.06	(1.01)	1.97
Volga	-0.21	(-2.90)	0.96	(10.85)	3.30
Vanna	-0.10	(-5.45)	-0.31	(-9.70)	5.47

At each date and for each risk dimension, we perform a cross-sectional regression of the market price of risk estimates against the stock return correlation between the underlying stock and the stock market portfolio. Entries report the time-series averages and their Newey-West t values (in parentheses) of the cross-sectional regression intercept, slope, and R^2 estimates.

since the market price of risk estimates are constructed per unit risk of each risk source, we use the scale-free correlation measure as the explanatory variable to capture the proportion of systematic return risk contribution to the stock.

Table 5 reports the time-series averages and the Newey and West (1987) t -values (in parentheses) of the regression intercept, slope, and R^2 estimates. The average R^2 estimates are low, ranging from 1.43% for the vega portfolios to 5.47% for the vanna portfolios; nevertheless, the average slope coefficients are statistically significant for three of four risk dimensions.

The average slope estimate on the vega portfolios is positive and statistically significant, suggesting that the average market pricing of vega risk increases with the stock return correlation. The delta-hedged option return variation is mainly driven by the implied volatility variation through the vega exposure. As the return risk premium is positive on average, the positive volatility risk premium from writing options can come partially from the positive return risk premium through the negative return-volatility correlation on the systematic return component. This contribution increases as the proportion of systematic stock return variation increases.

The average slope estimate on the gamma portfolios is also positive but small and with low

statistical significance. By contrast, the average slope estimate on the volga portfolios is highly positive and strongly statistically significant, suggesting that the average market pricing of volga risk increases strongly with the stock return correlation. For the vanna portfolios, both the average intercept and the average slope are strongly negative. Therefore, not only the average market pricing of vanna risk is negative, but the market price estimates also become more negative for stocks with higher return correlations with the market portfolio.

The volatility feedback effect can be the channel for transmitting a positive systematic return risk premium to a negative volatility risk premium; nevertheless, under diffusive price dynamics, the transmission mechanism is less clear from a positive systematic return risk premium to a positive volga risk premium and a negative vanna risk premium. The observed transmission potentially asks for a price dynamics that include random price jumps. Under a stylized setting where the wealth process follows a pure jump Lévy process and the investors have constant relative risk aversion preferences, Polimenis (2006) shows that a positive relative risk aversion makes the risk-neutral return distribution more negatively skewed with higher risk-neutral variance and fourth cumulant. Calibrating a dampened power law process to the stock index return history and the stock index options behavior, Wu (2006) shows that the market charges little risk premium on upside market jumps, but charges the highest possible risk premium on downside market jumps. A large market pricing on downside market jumps makes the index option implied volatility skew more negatively sloped, and leads to a highly positive market pricing estimate on the index option volga risk and a highly negative market pricing estimate on the index option vanna risk. If the idiosyncratic downside risk of a particular company is not priced as heavily, one would observe the dependence structure documented in Table 5: The market pricing of volga risk increases with the stock return correlation whereas the market pricing of the vanna risk becomes more negative. The positive dependence of the market pricing of gamma risk on the stock return correlation, although

not as statistically significant, is also consistent with such a positive market pricing hypothesis on the downside market jump risk.

5.2. Systematic versus idiosyncratic option return risk

In principle, the argument of the classic capital asset pricing theory is applicable to any securities market with frictionless trading. If the options transactions were costless, the classic argument would also hold for the delta-hedged excess returns on both individual option contracts and the risk-targeting option portfolios. However, the trading costs on stock options are magnitudes larger than the trading costs on the underlying stocks. Diversification through forming a market portfolio of options is anything but cost free. Therefore, it remains an unresolved but important question on how applicable the classic theory is to the options market.

To examine the implication of the classic theory on delta-hedged option returns, along each risk dimension, we form an equal-weighted market portfolio on the corresponding risk-targeting option portfolios. Starting in January 1998, at each month, we estimate the excess return correlation between the risk-targeting option portfolio on each stock and the corresponding market portfolio of the risk-targeting portfolios with a one-year rolling window. We use this option return correlation estimate to capture the proportion of systematic delta-hedged option return variation in each risk-targeting option portfolio.

We perform a cross-sectional regression of the market pricing estimates against this option return correlation estimate for each risk dimension. Table 6 summarizes the regression results. The time-series averages of the slope estimates are all statistically significant. The signs of the slope estimates are largely driven by the signs of the average market pricing estimates on the corresponding market portfolio. When the average market pricing estimates of the market portfolios are positive,

Table 6
Market price of risk variation with systematic option return risk

Risk	Intercept		Slope		R^2 , %
Vega	0.20	(2.68)	-0.23	(-1.82)	1.77
Gamma	0.03	(0.92)	0.04	(2.13)	0.83
Volga	0.13	(1.74)	0.45	(9.38)	3.66
Vanna	-0.24	(-13.73)	-0.04	(-3.68)	1.70

At each date and for each risk dimension, we perform a cross-sectional regression of the market price of risk estimates against the option excess return correlation between the risk-targeting option portfolio of a stock and the corresponding market portfolio of option portfolios. Entries report the time-series averages and their Newey-West t values (in parentheses) of the cross-sectional regression intercept, slope, and R^2 estimates.

such as for the gamma and volga portfolios, the market pricing estimates of individual stock option portfolio increase with the option return correlation estimates. When the average market pricing estimates of the market portfolios are negative, such as for the vega and more so for the vanna portfolios, the direction of dependence becomes negative.

Nevertheless, in all cases, the average R^2 estimates from these cross-sectional regressions are low, suggesting that a large proportion of the market pricing variation cannot be explained by the classic asset pricing theory. These remaining variations are either driven by some systematic but unknown risk sources, or reflect idiosyncratic pricing variations driven by temporary supply and demand shocks. Regardless of the underlying source, the analysis in the previous section shows that the market pricing variations are highly informative and can predict the ex post realized excess returns of the corresponding risk-targeting option portfolios in accordance with the expectation hypothesis. While investing the equal-weighted market portfolios of the risk-targeting option portfolios generates low average excess returns with different signs across different risk dimensions, long-short allocations to the risk-targeting option portfolios in proportion to the market price of risk estimates lead to high information ratios across all four risk dimensions.

6. Concluding remarks

Researchers are deeply interested in the risk-return behaviors of option investments, yet investment analysis in options is confounded by the multi-dimensional nature of option contracts. Each stock can have different number of option contracts across different strikes and maturities. These contracts have varying degrees of exposures to several sources of risks, making comparisons across different stocks difficult. Historically, empirical research often focuses on a single option contract or combination per each stock at similar maturity and moneyness so that the chosen option contracts have similar exposures and researchers can focus on the cross-stock variation in the underlying risk magnitude.

In this paper, we propose a theoretical framework to consolidate the information in all the option contracts on a stock into four risk-targeting option portfolios. Each portfolio targets a unit risk exposure to a particular risk source and zero exposures to other risk sources. The four portfolios explain about 99% of the cross-sectional option pricing variation and about 80% of the option excess return variation over the next month. As such, they can summarize the option pricing and return behaviors of all the option contracts with little information loss.

The theoretical construct not only performs dimension reduction effectively, but also offers an important dimension absent from the literature: A simple pricing regression identifies the market price of risk on the four risk dimensions. The market price of risk estimates strongly and positively predict the future excess returns of the corresponding risk-targeting option portfolio in line with the expectation hypothesis.

To highlight the general applicability of the theoretical construct, this paper applies minimum filtering on stock and option contract selection, and performs the empirical implementation with the simplest assumptions possible, leaving much to future research. First, accurately identifying

the market price of risk requires accurate statistical forecast on the corresponding risk. This paper uses one-year historical moment estimators to create a simple and conservative benchmark for the statistical forecasts. Future research can lever the large literature on variance forecasting to generate more refined variance and covariance forecasts on the stock return and implied volatility changes. More accurate moment forecasts shall lead to more accurate portfolio construction and more accurate market price of risk estimates, which shall ultimately lead to more profitable long-short portfolio option portfolio constructions.

Second, we assume no predictability in the directional implied volatility movement in extracting the market price of vega risk. As vega risk is the major source of variation for delta-hedged option investments, predicting the directional implied volatility movement of a fixed option contract, or the whole implied volatility surface, is a complex but highly important task. Accurately accounting for the market expectation on the directional implied volatility movement can enhance the accuracy of the market price of vega risk estimates and can potentially mitigate the observed biases in the expectation hypothesis regression.

Third, the BMS representation is a good starting point and is largely in line with practical risk considerations, but our second-order expansion relies on the assumption of small price and implied volatility moves. When price jumps become prominent, the BMS representation may become inappropriate or even unstable, especially for short-dated options (Carr and Wu (2003)). Future research can explore alternative representations that can better capture the risk exposures of short-dated option contracts facing potentially large random jumps, either with random arrival time or upon pre-schedule events.

Finally, much work is needed, both theoretical and empirical, to better understand the structural drivers of the cross-sectional and time-series variations of the market price of risk estimates.

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